

Errata for *Introductory Lectures on Equivariant Cohomology*

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Thanks go to Woohyeok Jo who found about a dozen of these errors.

- p. 7, line 5: “containing $f(y)$ ” should be “containing $f(x)$ ”.
- p. 22, bottom line: “ $h \in G$ ” should be “ $h \in H$ ”.
- p. 23, line 2 of Example 3.5: “ $\mathbb{C}^n$ ” should be “ $\mathbb{C}^n - \{0\}$ ”.
- p. 25, third line of the proof: the target of ϕ should be h^*E instead of E .
- p. 29, last sentence before §4.1: “abbreviate” should be “abbreviate”.
- p. 40, Example 5.5: “principle” should be “principal”.
- p. 83, eq. (10.1): the initial $-$ sign should be $+$.
- p. 107, proof of Cor. 13.8, line 1: “the lemma” should be “Proposition 13.6”.
- p. 111, line following fourth display: Delete the extra “the” in the sentence “ $F(h, a)$ is continuous on its domain”
- p. 119, Th. 14.5: Rewrite the hypothesis as “If $\omega, \tau \in \dots$ are homogeneous”.
- p. 119, third line of the proof: $\omega^{k\ell}$ should be $\tau^{k\ell}$.
- p. 123, first display: “ $\sum_{i,j}$ ” should be “ $\sum_k$ ”.
- p. 132, Prop. 16.8: The proposition should end with a period, not a comma.
- p. 136, line 3 of first display: “ $\sum_{i,j}$ ” should be “ $\sum_{i,j,k}$ ”.
- p. 137, line 3 of first display: “ $\sum_{i,j}$ ” should be “ $\sum_{i,j,k}$ ”.
- p. 137, Lemma 17.3, line 1: “horizontal distribution” should be “right-invariant horizontal distribution”. In the second to last line of the proof, change “a horizontal” to “the horizontal”.
- p. 148, proof of Prop. 18.8, first line: Delete of the repeating “to”s.
- p. 156, last display above §19.4: “ x_i ” should be “ X_i ”.
- p. 157, proof of Th. 19.2, add a new paragraph at the beginning: Let $W^{p,q} = \bigwedge^p(\mathfrak{g}^\vee) \otimes S^q(\mathfrak{g}^\vee)$. In this proof, we define the temporary degree of an element of $W^{p,q}$ to be $p + q$, instead of the usual $p + 2q$, and we define W^m to be $\bigoplus_{p+q=m} W^{p,q}$. Then W^0 is the same whether with respect to the usual degree or the temporary degree and $W^+ = \bigoplus_{m>0} W^m$ is also the same, so that if the statement of the theorem is true with respect to the temporary degree, it is also true with respect to the usual degree. The Weil differential δ and the cochain homotopy K will both have temporary degree 0, but this is fine. Cohomology is still defined to be $\ker \delta / \text{im } \delta$. In the following proof, the “degree” means the “temporary degree.”

- p. 157, first line of third paragraph: Change “degree +1 and -1 respectively” to “degree 0”.
- p. 157, bottom line: “ $W\mathfrak{g}$ ” should be “ $W(\mathfrak{g})$ ”.
- p. 159, third paragraph, line 3: $\partial = \delta \otimes 1 + (-1)^{\deg \alpha} 1 \otimes d$ on $\alpha \otimes \omega$
- p. 169, in (5) after equation (21.4): “ $H_i(\theta_i) = 0$ ” should be “ $H(\theta_i) = 0$ ”.
- p. 170, line 2 in (3): Delete “and $\iota_i^2 = 0$ ”.
- p. 172, Proposition 21.2 (i), line 2: “degree -1 ” should be “degree $+1$ ”.
- p. 172, Proof of Proposition 21.2 (i), line 2: “degree -1 ” should be “degree $+1$ ”.
- p. 173, the line above Definition 21.3: $\mathfrak{g}^\vee \rightarrow V$ should be $\mathfrak{g} \rightarrow V$.
- p. 175, Section 21.6, line 2: “adjoint” should be “coadjoint”.
- p. 175, bottom display: “ $u_n^{i_\ell}$ ” should be “ $u_\ell^{i_\ell}$ ”.
- p. 189, Definition 23.1, first line: “ring” should be “commutative monoid (under addition)”, since $\mathbb{N}$ is not a ring.
- p. 198, Example 24.4, last line: “not free” should be “not free for $n > 1$ ”.
- p. 200, Proposition 24.8, line 2: “acts on a manifold” should be “acts locally freely on a manifold”. We need to assume a locally free action so that the fundamental vector field $\underline{X}$ would be nowhere-vanishing (Proposition 24.5).
- p. 208, fourth line: $(TM)|_s$ here the subscript “ s ” should be written in capital S (similar typos are also in [p.218, last line] and [p.232, last displayed equation]).
- p. 211, Proposition 25.13(iv): “if U is an open set” should be “if U is an invariant open set”.
- p. 236, Third paragraph, first line should read: “Suppose $f: M \rightarrow M$ is a C^∞ map with isolated fixed points and does not map a non-fixed point to a fixed point.” It is not possible to define the image of a non-fixed point under $\tilde{f}$.
- p. 246, third line: “Lemma 21.5” should be “Proposition 21.5”.
- p. 255, line after first display: “ $\sum m_r$ ” should be “ $\sum r m_r$ ”.
- p. 257, commutative diagram in display: $H^{n-q}(M)$ should be $H^{m-q}(M)$.
- p. 261, second displayed equation: there is an extra left bracket “(” before “ $S(\mathfrak{g}^\vee)$ ”.
- p. 264, first displayed equation: $\delta = \delta \otimes 1 + (-1)^{\deg \alpha} 1 \otimes d$ on $\alpha \otimes \omega$.
- p. 271, line 4 of proof of Theorem A.7(a): as a universal G -bundle, we should take $V(k, \infty) \rightarrow V(k, \infty)/G$ instead of $V(k, \infty) \rightarrow G(k, \infty)$.
- p. 290, solution to Problem 7.1, first display: “ $F_G([e, m])$ ” should be “ $F_G([e, m], t)$ ”.
- p. 290, second to last line : “ $f \circ g$ ” should be “ $f \circ h$ ”.
- p. 292, solution to Problem 11.3, line 3 of display: “ e^{-tA} ” should be “ e^{-tA} ”.
- p. 293, second to last line: “ $(\sum u^I +$ ” should be $\sum (u^I +$.
- p. 294, solution to Problem 24.1, first displayed equation: a right bracket $)$ is missing before the period.